

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
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Driven by a problem solving mindset and fueled by curiosity, I aspire to develop genuine solutions for the real-world, impactful problems.

SKILLS

Languages: English (C1 [IELTS 8.0](#) )
Programming: Python, Julia, MATLAB, C/C++, R, SQL
Tools: Linux, Git, Bash, PyTorch, OpenCV, open3d, Docker, Quarto, SSH, L^AT_EX

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering Budapest, Hungary
Supervisor: [Márton Vaitkus](#) 
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics Ulaanbaatar, Mongolia
GPA: 3.2/4.0
Supervisor: [Galtbayar Artbazar](#) 
Thesis: [Evaluating land degradation using image processing methods](#) 

EXPERIENCE

Center of Mathematics Application, NUM Oct, 2023 – June, 2025
Undergraduate Research Intern

- Contributed to the ongoing research effort by developing a [drone imagery analysis pipeline](#)  which segments the drone images into vegetation, bare soil, and quantifying presense of weed in the vegetation. The aim of the research project was to reduce reliance on manual field surveys by efficiently analyzing the drone images, this potentially saving hundreds of man-hours of monitoring in Mongolian steppe.
- Gained hands-on experience with drone imagery analysis and AI-based computer vision models in PyTorch. Performed drone image thresholding and segmentation using MATLAB and Julia. Worked within a Linux environment, configured and maintained secure remote access to lab servers via SSH and Tailscale.
- Architected a unified, documented data storage structure for research backups, directly improving data retrieval efficiency and establishing a data management guidelines for future students.
- Engaged in field survey data collection and multispectral measurements; gained first-hand insight into ecological data collection to better design the drone imagery analysis pipeline with domain-specific knowledge.

PROJECTS

[aiSim LiDAR point cloud sensor fusion for 3D Gaussian Splatting](#)

- Initiated a project to train 3D Gaussian Splatting model from autonomous driving simulation data, under the supervision of Professor Márton Vaitkus. Achieved colorization the LiDAR point cloud measurements obtained from [aiSim](#)  simulator using the surround camera images.
- Engineered the transformation between 3D world to camera sensor coordinate frame, made the observations into a consistent coordinate convention by extensive testing and keeping track of each 3D data processing softwares.

[Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Reproduced machine learning demonstrations from Bishop & Bishop's *Deep Learning Foundations and Concepts* using Julia notebooks with innovative interactive, reactivite programming environment using the [Pluto.jl](#)  package.

AWARDS

Stipendium Hungaricum Scholarship Sep, 2025